

Element E

STEM Elements Present in the Solution

Science - This problem deals with issues with the temperature and humidity in the art room. In chemistry, we learn about temperature and humidity, and we can use that knowledge to combat the problem. The student engineers applied their understanding of meteorological concepts, when studying the effects of humidity on paper.

Technology - The design team used an online computer aided design (CAD) website called tinkercad to help with the designing of the final prototype. They plan on using a 3D printer and laser cutters to make the final design. The student engineers used their researching skills, to browse the internet when researching the effects of humidity in tight spaces.

Engineering - The construction of the solution will require engineering because the designing of the prototype will have to undergo a design process called the “engineering design process”. The building process will require engineering, planning, building, communication, and maintenance as needed before the final product. All of these skills are key for engineers to successfully carry out a project.

Math - The measurements of the design will require some calculations to be made to ensure the design is the right length, width, & height. The student engineers applied ratios to scale down their original design.

Limitations & Impact In STEM

The student engineers will want to focus heavily on one particular area of stem (such as math) instead of equally concentrating on all the areas of STEM. This can cause the design to deter away from its original purpose, which is to ensure that humidity would not affect the paper and clay in the art teacher's room. If the students focus too much on the technological aspects of the design, then it is likely that the functionality and the math proportions of the design decreases gradually. The same is true for student engineers focusing solely on any other STEM element. However, when the student engineers focus on one stem element alone, they are able to stay on track and meet all the criteria and constraints that might result from that STEM element. They will be able to weigh their pros and cons, without feeling too much of an impact from other STEM elements.

Justifying the Solution

To justify the solution to this problem the student engineers will observe and document how many days the paper and clay can stay dry from the humidity in the art room. This will then be used to test how durable the design is, which allows the engineers to make adjustments along the way.